STEADY STATE ERROR OF FEEDBACK CONTROL SYSTEMS

One of the fundamental reasons for using feedback is the improvement in the reduction of steady stare error of the system. The steady state error is the error after the transient response has decayed (i.e., the value of error when t→ ∞. Let us compare the steady state errors of an open loop and closed loop system.

Consider the open loop system shown in figure.

The input is $X(s)$ and the output is $Y(s)$. The system transfer function is $G(s)$.

$$Y(s) = G(s).X(s)$$

The diffrence between the desired output (input) and the actual output is known as the error

$$e(t) = x(t) - y(t)$$

$E_o(s) = X(s) - Y(s) = X(s) - G(s)X(s)$ (the suffix 'o' denotes open loop and 'c' closed loop)

$E_o(s) = [1 - G(s)]\, X(s)$ .... (1)

For a unit step input, $X(s) = 1/s$

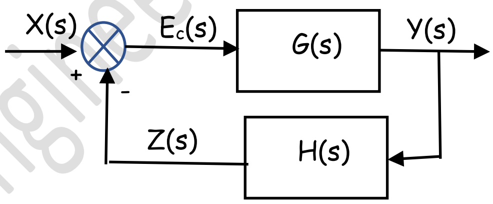
The steady state error $= \lim_{t\to\infty} e(t) = \lim_{s\to 0} s\, E_0(s) = \lim_{s\to 0} s\,[1 - \mathrm{G(s)}]\frac{1}{s} = 1 - G(0)$ .... (2)

Consider the closed loop system

$Z(s) = Y(s)\, H(s)$ and $Y(s) = E_c(s).G(s)Ec(s) =$ $X(s) - Z(s) = X(s) - Y(s)H(s) = X(s) - E_c(s)G(s)H(s)$

$$E_c(s)\,\{1 + G(s)H(s)\} = X(s)$$

$$E_c(s) = \frac{1}{1+G(s)H(s)}\, X(s) \quad \text{......................(3)}$$



Assuming a unity feed back, $H(s) = 1$

$$E_c(s) = \frac{1}{1+G(s)}\, X(s) \quad \text{................................(4)}$$

The steady state error for a unit step input ($X(s) = 1/s$)

$$= \lim_{t\to\infty} e(t) = \lim_{s\to 0} s\, E_c(s) = \lim_{s\to 0} s\left[\frac{1}{1+G(s)}\right]\frac{1}{s} = \frac{1}{1+G(0)} \quad \text{..(5)}$$

The value of $G(s)$ when $s = 0$ is often called the dc gain and normally is greter than 1. Therefore, comparing equations (2) and (3), it can be seen that the steady state error of a closed loop system is considerably less than the open loop system.

We can now determine the steady state error of the system for three standard test inputs

$$E(s) = \frac{1}{1+G(s)H(s)}\, X(s)$$

Remember $G(s)$ stands for the open loop transfer function (transfer function of the forward path)

In order to explain how these test signals are used, let us assume a position control system, where the output position follows the input commanded position.

Step inputs represent constant position and thus are useful in determining the ability of the control system to position itself with respect to a stationary target. An antenna position control is an example of a system that can be tested for accuracy using step inputs.

Ramp inputs represent constant-velocity inputs to a position control system by their linearly increasing amplitude. These waveforms can be used to test a system's ability to follow a linearly increasing input or, equivalently, to track a constant velocity target. For example, a position control system that tracks a satellite that moves across the sky at a constant angular velocity.

Parabolic inputs, whose second derivatives are constant, represent constant acceleration inputs to position control systems and can be used to represent accelerating targets, such as a missile.

STEP INPUT (constant)

The input is a step of magnitude A, $x(t) = A$

$$X(s) = \frac{A}{s}$$

$$E(s) = \frac{1}{1+G(s)H(s)}\, X(s) = \frac{1}{1+G(s)H(s)} \cdot \frac{A}{s}$$

$e_{ss} = \lim_{s\to 0} s.E(s) = \lim_{s\to 0} s.\left[\frac{1}{1+G(s)H(s)}\right] . \frac{A}{s} = \frac{A}{1+\lim_{s\to 0} G(s)H(s)} = \frac{A}{1+K_p}$ where $K_p = \lim_{s\to 0} G(s)H(s)$ and is known as position error constant

The general form of the forward transfer function is $G(s) = K\frac{\prod_{i=1}^{M}(s+z_i)}{\prod_{k=1}^{P}(s+p_k)}$ where $\prod$ denotes the product of factors. If there are N poles at the origin, then we can decompose the denominator as $s^N \prod_{k=1}^{Q}(s+p_k)$ so that $N+Q = P$.

$G(s) = K\frac{\prod_{i=1}^{M}(s+z_i)}{s^N\prod_{k=1}^{Q}(s+p_k)}$

for example $G(s) = 100\frac{(s+2)(s+1)}{s^3(s+3)(s+10)}$

The number of s in the denominator indicates number of integrations. The number of inegrations N is often called type number.

For a type zero system, $N = 0$, $G(s) = K\frac{\prod_{i=1}^{M}(s+z_i)}{\prod_{k=1}^{Q}(s+p_k)}$

for example $G(s) = 100\frac{(s+1)}{(s+3)(s+10)}$

$e_{ss} = \frac{A}{1+K_p}$ where $K_p = \lim_{s\to 0} G(s)H(s).$

If $N \geq 1$, $G(s) = K\frac{\prod_{i=1}^{M}(s+z_i)}{s^N\prod_{k=1}^{Q}(s+p_k)}$, $\mathrm{Kp} = \lim_{s\to 0} G(s)H(s). = \infty$

and steady stae error, $e_{ss} = \frac{A}{1+K_p} = 0$

RAMP INPUT (linear)

Input is a ramp with slope A, $x(t) = At$

$$X(s) = A/s^2$$

$$e_{ss} = \lim_{s\to 0} s.E(s) = \lim_{s\to 0} s.\frac{1}{1+G(s)H(s)}.\frac{A}{s^2} = \lim_{s\to 0}\frac{A}{s+sG(s)H(s)} = \frac{A}{\lim_{s\to 0} sG(s)H(s)} = \frac{A}{K_v}$$

Where $K_v = \lim_{s\to 0} \mathrm{sG(s)H(s)}$ and is known as velocity error constant

If $G(s)$ is type 0 (no s term in the denominator), $G(s) = K\frac{\prod_{i=1}^{M}(s+z_i)}{\prod_{k=1}^{Q}(s+p_k)}$

$K_v = \lim_{s\to 0} \mathrm{sG(s)H(s)} = 0$ >>> $e_{ss} = \infty$. A type 0 system can not track a ramp input.

If $G(s)$ is type 1 (one s in the denominator), $G(s) = K\frac{\prod_{i=1}^{M}(s+z_i)}{s\prod_{k=1}^{Q}(s+p_k)}$

$K_v = \lim_{s\to 0} \mathrm{sG(s)H(s)} = constant$ >>> >>> then we have a constant steady state error $e_{ss} = \frac{A}{K_v}$

If $G(s)$ is type 2 or higher (2 or more s in the denominator), $G(s) = K\frac{\prod_{i=1}^{M}(s+z_i)}{s^N\prod_{k=1}^{Q}(s+p_k)}$

$K_v = \lim_{s\to 0} \mathrm{sG(s)H(s)} = \infty$ then, $e_{ss} = 0$

ACCELERATION INPUT (Parabolic)

If the input is $x(t) = \frac{1}{2}At^2$, then $X(s) = \frac{A}{s^3}$

$$e_{ss} = \lim_{s\to 0} s.E(s) = \lim_{s\to 0} s.\frac{1}{1+G(s)H(s)}.\frac{A}{s^3} = \lim_{s\to 0}\frac{A}{s^2+s^2G(s)H(S)} = \frac{A}{\lim_{s\to 0} s^2G(s)H(s)} = \frac{A}{K_a}$$

Where $K_a = \lim_{s\to 0} \mathrm{s^2G(s)H(s)}$ and is known as acceleration error constant

For type 0 *and* 1 systems, $K_a = \lim_{s\to 0} \mathrm{s^2G(s)H(s)} = 0$ or $e_{ss} = \infty$

For type 2 systems, $K_a = \lim_{s\to 0} \mathrm{s^2G(s)H(s)} = constant$, e_{ss} is constant and is equal to $\frac{A}{K_a}$

For type 3 and above, $K_a = \lim_{s\to 0} \mathrm{s^2G(s)H(s)} = \infty$, $e_{ss} = 0$

We can summarise this as follows

Type Number / Input	0	1	2	3
Step, $x(t) = A$, $X(s) = A/s$	$e_{ss} = \frac{A}{1+K_P}$	0	0	0
Ramp, $x(t) = At$, $X(s) = A/s2$	∞	$e_{ss} = \frac{A}{K_v}$	0	0
Parabolic, $x(t) = \frac{1}{2}At^2$ $X(s) = A/s3$	∞	∞	$e_{ss} = \frac{A}{K_a}$	0

The error constants, K_p, K_v , and K_a, of a control system describe the ability of a system to reduce or eliminate the steady state error. They are used as numerical measures of steady state performance. The system designer determines the errror constants for a given system and explores the methods of increasing the error constants (thereby minimising the steady state error) while maintaing an acceptable transient response.

We can reduce the feedback system shown in the figure in to a single block of transfer function $T(s) = \frac{G(s)}{1+G(s)H(s)}$ with input $X(s)$ and output $Y(s)$. Then $Y(s)$ can be written as

$$Y(s) = T(s).X(s)$$

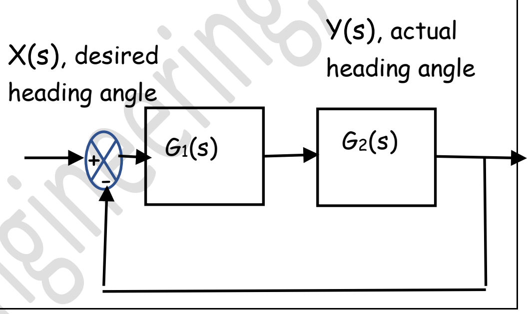
The error $E(s) = \textit{desired output} - \textit{actual output} = X(s) - Y(s) = X(s)\{1 - T(s)\}$

The steady state error, $e_{ss} = \lim_{s\to 0} s\,E(s) = \lim_{s\to 0} s\,X(s)\{1 - T(s)\}$ (Final value theorem)

This can also be used to find out the steady state error.

Example 1

A severely disabled person could use a mobile robot as an assisting device. The steering control system for such a robot can be represented by the block diagram. The steering controller $G_1(s) = K_1 + \frac{K_2}{s}$. The vehicle dynamics is represented by $G_2(s) = \frac{K}{1+Ts}$. Determine the steady state error for a step and ramp input.



The open loop transfer function of the given system

$G(s) = G_1G_2 = (K_1 + \frac{K_2}{s})\left(\frac{K}{1+Ts}\right) = K.\frac{sK_1+K_2}{s\,(1+Ts)}$. Look at the denominator, this is type 1 system

$$H(s) = 1$$

The position error constant for a step input $x(t) = A$,

$K_p = \lim_{s\to 0} G(s)H(s). = \lim_{s\to 0} K.\frac{sK_1+K_2}{s\,(1+Ts)} = \infty.$

The steady state error $e_{ss} = \frac{A}{1+K_p} = 0$

You can also verify the result using $e_{ss} = \lim_{s\to 0} s\,X(s)\,\{1 - T(s)\}$

If the input is ramp, $x(t) = At$

The velocity error constant $K_v = \lim_{s\to 0} sG(s)H(s) = \lim_{s\to 0} s\,K.\frac{sK_1+K_2}{s\,(1+Ts)}$.

$K_v = \lim_{s\to 0} s\left(\frac{sK_1+K_2}{s}\right)\left(\frac{K}{1+Ts}\right) = K.K_2$

The steady state error, $e_{ss} = \frac{A}{K_v} = \frac{A}{K.K_2}$

Example 2

What is the steady state error to a constant reference input $r(t) = 3cm.1(t)$ for the following feedback positioning system?

$G(s) = \frac{10}{s+1}, R(s) = \frac{3}{s}, H(s) = 1$

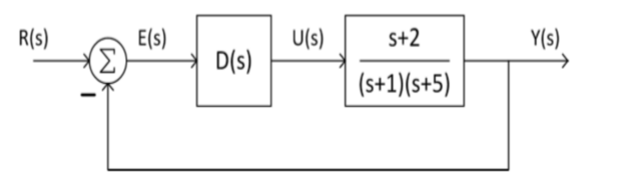
Steady state error for a step input $e_{ss} = \frac{A}{1+K_p}$ where $K_p = \lim_{s\to 0} G(s)H(s). = \lim_{s\to 0} \frac{10}{s+1} = 10$

$e_{ss} = \frac{A}{1+K_p} = \frac{3\ cm}{1+10} = 0.27\ cm$

You can also verify the result using $e_{ss} = \lim_{s\to 0} s\,R(s)\,\{1 - T(s)\}$

Example 3

Design the controller $D(s)$ for an error of 0.05 for a unit ramp input



Looking at the transfer function of the plant, we can understand that it is $type\ 0$. For getting a constant steady state error for a unit ramp, the forward path transfer function should be $type\ 1$. That is there should be one integrator in the forward path.

Assuming the simplest case, $D(s) = \frac{K}{s}$

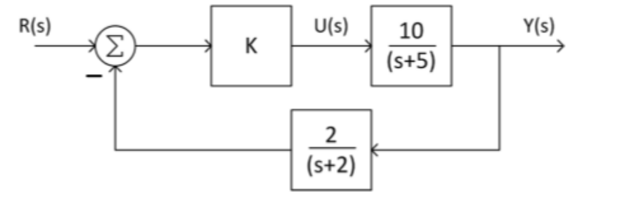
$G(s) = \frac{K\,(s+2)}{s(s+1)(s+5)}$, $H(s) = 1$, $R(s) = 1/s2$ $(A = 1)$, $e_{ss} = 0.05$

$e_{ss} = \frac{A}{K_v}$ where $K_v = \lim_{s\to 0} sG(s)H(s) = \lim_{s\to 0} s\frac{K\,(s+2)}{s(s+1)(s+5)} = \frac{2K}{5}$

i.e., $0.05 = \frac{1}{\frac{2K}{5}}$ which gives $K = 50$.

Example 4

Determine the controller gain K to limit the steady state error of the system to 0.02 for a constant reference input



Open loop transfer function $G(s) = \frac{10\,K}{s+5}$

$H(s) = \frac{2}{s+2}$, For a unit step input $R(s) = \frac{1}{s}$, $A = 1$

$e_{ss} = 0.02$

$e_{ss} = \frac{A}{1+K_p}$ where $K_p = \lim_{s\to 0} G(s)H(s). = \lim_{s\to 0} \left(\frac{10K}{s+5}\right)\left(\frac{2}{s+2}\right) = 2K$

i.e., $0.02 = \frac{1}{1+2K}$

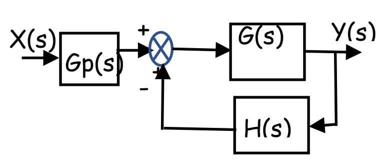
$1 + 2K = 50$ which will give thr Controller gain $K = 24.5$

You can also verify the result using $e_{ss} = \lim_{s\to 0} s\,R(s)\,\{1 - T(s)\}$

Exercise 19

19.1 A system with unity negative feed back has transfer function G(s) $= \frac{10(s+4)}{s(s+1)(s+2)(s+5)}$ Determine the steady state error for a unit step and unit ramp input.

19.2 A feedback system is shown in figure. $G(s) = \frac{K}{s(s+2)}$ and $H(s) = \frac{(s+3)}{(s+0.1)}$. Determine the steady state error for a unit step when $K = 0.4$ and $G_p(s) = 1$. Select an appropriate value for $G_p(s)$ sothat the steady state error becomes zero for a unit step input.



19.3 A thermometer requires $1\ min$ to indicate 98% of the response to a step input. Assming it as a first order system, determine the time constant. If the thermometer is placed in a bath whose temperature is changing linearly at a rate of $10\ deg/min$, how much error will it show?